Valuation: Part I

Relative Pricing

Relative Pricing

- Determine the price of an asset by comparing the market value of similar or comparable assets

- Important steps
  - Determine the comparable assets
  - Convert the market price of comparable assets into a ratio relative to the fundamentals
  - Acquire the pricing ratio by comparison
    - » Adjust all possible deviations

The Importance of Relative Pricing

- Relative pricing gains its popularity in the Wall Street

– Almost 85% of the stock research reports are based on the multiple of comparable stocks

– More than 50% of M&A pricing is based on multiples

– Multiples are usually the basis for making final investment decisions

- Why relative pricing

– Pricing is often a bargain.

Demands for Pricing

- On February 3, 2012, the director of the relevant departments of the CSRC claimed that the CSRC would study the supervision of the underwriting process, and will direct it to **rationally** set IPO prices. The CSRC is working on a measure to promote the fair pricing of new shares, which is to entrust a third party to publish the average P/E ratio of listed companies in various industries for investors' reference.
- IPO pricing is generally no more than 25% of the industry's P / E ratio. ⇒ (relative pricing)
- Since 2014, with a P/E ratio of 22.98 times, regulators almost ensured that every investor subscribing to new shares is profitable.

↳ maximum constrain

[actual PE ratio may > 30
∴ investors are willing to buy
if lucky to get oppurtunity, then gain money]

Understanding Multiples

- The definition of multiples
  - How to figure out the multiples
- Analyzing multiples
  - Know the distribution of multiples
- The factors of multiples
  - What determines the multiples?
- Using multiples
  - Determining comparable collections and adjusting deviations is not easy.

The Calculation of Multiples

Some important financial data

–Sales Revenue

» The first row of the Income Statement

–Gross Profit

» =Sales –COGS

» Gross Profit Margin =Gross Profit/Sales Revenue

– Earnings before Interest and Taxes（EBIT）

» =Gross Profit– Selling, general and administrative expenses – Depreciation and amortization–Other expenses

» EBIT Margin=EBIT/Sales Revenue

Revenue	
Sales Revenue	$20,438
Operating Expenses	
Cost of goods sold	$7,943
Selling, general and administrative expenses	$8,172
Depreciation and amortization	$960
Other expenses	$138
Total operating expenses	$17,213
Operating income	**$3,225**
Non-operating income	$130
Earnings before Interest and Taxes (EBIT)	**$3,355**
Financial income	$45
Income before Interest Expense (IBIE)	**$3,400**
Financial expense	$190
Earnings before income taxes	**$3,210**
Income taxes	$1,027
Net Income	$2,183

如中葡股份2016年实现营业收入2.65亿元，归属于上市公司股东的净利润0.13亿元，而其理财收益高达0.56亿元，占总营收的比例为21%

The Calculation of Multiples

- **Some important financial data**
  - Earnings before Interest, Taxes, Depreciation and Amortization (EBITDA)
    - » =EBIT + Depreciation and amortization (DA)
    - » EBITDA Margin=EBITDA/Sales Revenue
  - Net Income
    - » =EBIT – Interest– Taxes – Non-controlling Interest(NCI) – Preferred Dividend
    - » It is customary to calculate net income as earning per share(EPS)
    - » P/E ratio = Share price/ EPS
    - » ROE=Net Income/Net Asset
    - » ROA=Net Income/Total Asset

The Calculation of Multiples

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项　　目
一、营业收入
减：营业成本
营业税金及附加
销售费用
管理费用
财务费用
资产减值损失
加：公允价值变动收益（损失以“-”号填列）
投资收益（损失以“-”号填列）
其中：对联营企业和合营企业的投资收益
二、营业利润（亏损以“-”号填列）
加：营业外收入
减：营业外支出
其中：非流动资产处置损失
三、利润总额（亏损总额以“-”号填列）
减：所得税费用
四、净利润（净亏损以“-”号填列）
五、每股收益：

The Calculation of Multiples

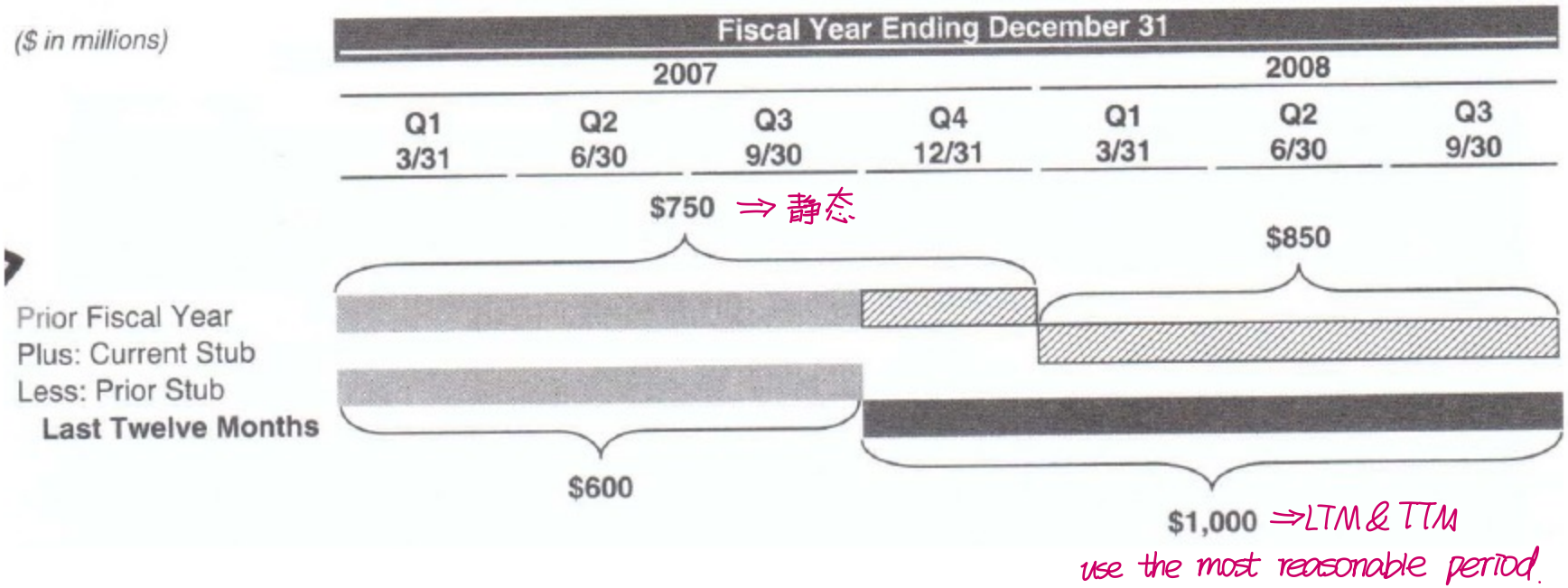
- Updating recent financial data.
  - Data for the past 12 months (LTM, TTM)
    - » = Data for last financial year+ Data for the current year – Corresponding data for last financial year

last 12 month

training 12 month

Update Recent Financial Data

EXHIBIT 1.25 Calculation of LTM 9/30/08 Sales



use the most reasonable period.

「Dynamic PE: can make PE smaller ⇒ company is cheap
(tend to use large earning) ⇓ time to buy
「have room to forecast」

静态 large PE: CEO incentive for higher salary

The Calculation of Multiple

- Adjust the income statement
  - adjust the extraordinary items in the income statement
  - example of expense
    - » caused by reorganization event → one-time
      - store closures, layoffs 解雇
    - » loss on sale of assets one time gain/loss
    - » changes in accounting rules eg. 折旧加速. delay tax expense.
    - » inventory cancellation
    - » derogation (损毁) of goodwill
    - » litigation (诉讼) loss

– there are "non-recurring," "extra," and "disposable" in the annual and quarterly reports. challenge whether they are truely one-time things.

Adjust the income statement

EXHIBIT 1.29 Adjusted Income Statement

($ in millions, except per share data)

Income Statement	Reported 2007	Adjustments +	Adjustments −	Adjusted 2007
Sales	**$1,000.0**			**$1,000.0**
Cost of Goods Sold	625.0		(5.0)	620.0
Gross Profit	**$375.0**			**$380.0**
Selling, General & Administrative	230.0			230.0
Restructuring Charges	10.0		(10.0)	-
Operating Income (EBIT)	**$135.0**			**$150.0**
Interest Expense	35.0			35.0
Pre-tax Income	**$100.0**			**$115.0**
Income Taxes	40.0	6.0		46.0
Net Income	**$60.0**			**$69.0**
Operating Income (EBIT)	$135.0	15.0		$150.0
Depreciation & Amortization	50.0			50.0
EBITDA	**$185.0**			**$200.0**
Weighted Avg. Diluted Shares	30.0			30.0
Diluted EPS	$2.00			$2.30

Inventory write-down

Restructuring charge related to severance from downsizing the sales force

= (Inventory write-down + Restructuring charge) x Marginal Tax Rate
= ($5.0 million + $10.0 million) x 40.0%

D&A is typically sourced from a company's cash flow statement although it is sometimes broken out on the income statement

$15.0 million add-back of total non-recurring items

The Calculation of Multiple

- Multiples
  - P/E ratio
    - » =share price/earnings per share
  - P/B ratio
    - » =share price/Book value per share
  - enterprise value and EBITDA
    - » =EV/EBITDA
  - enterprise value and EBIT
    - » =EV/EBIT
  - enterprise value and sales
    - » =EV/Sales

market value ↖ ↗ foundamental of company

not good as PE (Investors cares about earning, not book value)

if negative: useless. (should drop).
if earning → 0: also not useful. then can use PB ratio

real cash flow

firm value cash flow

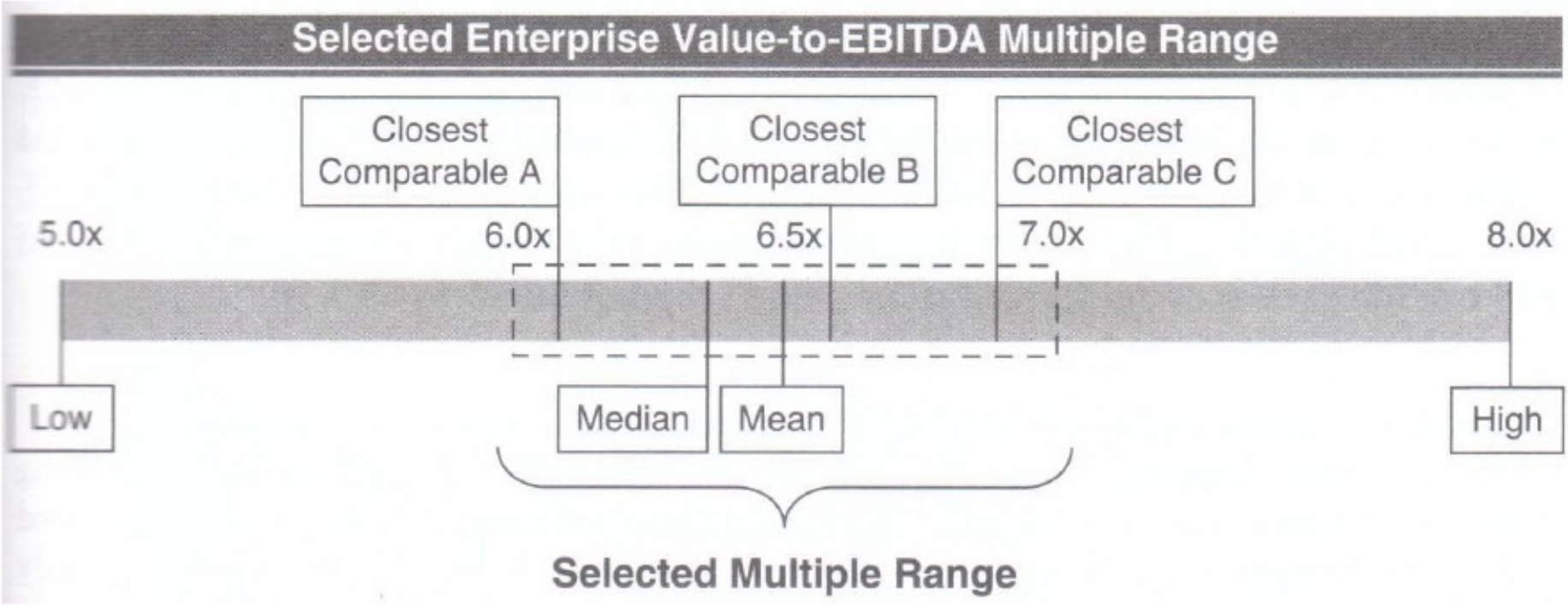
Top line in income statement.

Earning can do lots of adjustment
sales is more difficult to 改 (more objective)

Benchmarking

- Choose a comparable company
  - calculate key financial ratios and multiples
  - calculate the range of financial ratios and multiples of comparable company
    - » average, median, highest and lowest
    - » investigate and understand differences
- Benchmarking 对标
  - Establish the range of valuation
    - » in general, to find the mean and median of the most relevant industries

Benchmarking



Benchmarking

($ in millions, except per share data)

EBITDA	Financial Metric	Multiple Range	Implied Enterprise Value	Less: Net Debt	Implied Equity Value	Fully Diluted Shares	Implied Share Price
LTM	$200	6.50x – 7.50x	$1,300 – $1,500	(500)	$800 – $1,000	100	$8.00 – $10.00
2008E	215	6.00x – 7.00x	1,290 – 1,505	(500)	790 – 1,005	100	$7.90 – $10.05
2009E	230	5.50x – 6.50x	1,265 – 1,495	(500)	765 – 995	100	$7.65 – $9.95

Benchmarking

EXHIBIT 1.36 Valuation Implied by P/E – Share Price

($ in millions, except per share data)

Net Income	Financial Metric	Multiple Range	Implied Equity Value	Fully Diluted Shares	Implied Share Price
LTM	$70	12.00x – 15.00x	$840 – $1,050	100	$8.40 – $10.50
2008E	75	11.00x – 14.00x	825 – 1,050	100	$8.25 – $10.50
2009E	80	10.00x – 13.00x	800 – 1,040	100	$8.00 – $10.40

Benchmarking

($ in millions)

Net Income	Financial Metric	Multiple Range	Implied Equity Value	Plus: Net Debt	Implied Enterprise Value
LTM	$70	12.00x – 15.00x	$840 – $1,050	500	$1,340 – $1,550
2008E	75	11.00x – 14.00x	825 – 1,050	500	1,325 – 1,550
2009E	80	10.00x – 13.00x	800 – 1,040	500	1,300 – 1,540

Summary

① market valuation is correct.
② underlying company is same as these comparable company.

The advantages of relative pricing

– Utilization of matket information
  » Considering market growth, risk estimates, and market sentiment

↑ if over-pes消极, then our valuation also suffer over-pes~.

– Relativity
  » Easy to be measured and compared with other companies
– Fast and convenient
  » Easy to build a computing platform
– In time
  » Market information: updated every day

Summary

- The disadvantages of relative pricing
  - Utilize market information
    - » Market sentiment can lead to pricing bias.



Summary

The disadvantages of relative pricing

- Lack of comparable companies
  - » Baidu

PE really depends on growth rate.
(Baidu PE should >> Google) But use same PE to do IPO.
stock price tripled in 3 years (说明IPO价低了.
李Yanhong很伤心).

- The price could be very different from DCF.

Walren Buffet never believe relative price (no insight)

- Corporate individual factors

Precedent Transaction Cases Analysis

- Pricing by using multiples of comparable companies in previous M&A cases to price.

- It is a challenge to find comparable companies
  – Sometimes different industries can be considered when there is similar:
    - » Final product market
    - » Distribution channels
    - » Financial characteristics

Google bought Motorola Mobility LLC

- August 15, 2011
  - Google bought MOTOROLA with $12.5 billion
  - 63% premium *much higher than market value.*
  - How did they figure out $12.5 billion?



Google bought Motorola Mobility LLC

- MOTOROLA has 24,500 patents.
- In October 2010, Novel sold 882 patents that are worth $450 million.
  – How much is each patent worth?
- If Google buys 24,500 patents for $12.5 billion.
  – How much is each patent worth?
- Conclusion [the key is to get patents]
  – Google bought 24,500 patents plus a free mobile phone company.

How Are Patents Priced?

- Pricing of Kodak
  - Market value: On November 24, 2011, $310 million.
  - There are 1000 patents
  - If each patent can be sold for $510,204 → calculated in Novel case
    - » $510 million

How Are Patents Priced?

- **[Kodak's proposed $525 million patent sale plan was approved by a bankruptcy judge]**

  On January 11th, kodak proposed a price of $525 million to Intellectual.

  Proposal for Ventures and RPX to sell digital imaging patents was approved by bankruptcy judges, which made the company closer to coming out of bankruptcy protection.

Precedent Transaction Cases Analysis

- Steps
  - Select relevant M&A comparable companies
  - Collect relevant M&A and financial information
  - Calculate key data, ratios, and transaction multiples
  - Benchmark related mergers and acquisitions
  - Pricing

Precedent Transaction Cases Analysis

- Transaction details
  - Is the buyer a strategic investor or a financial investor?
    - » Strategic investors are generally willing to pay higher prices
  - What are the motivations of buyers and sellers?
    - » Is the seller in urgent need of cash?
  - Transaction procedures and nature
    - » The auction
    - » A hostile takeover 去 secondary market buy many stock. [- 股 valuation is much higher]
    - » Equal participation
  - Buying patterns
    - » Cash seller want cash. buyer: stock + cash
    - » Stock ⇒ usually cost more
      - ↳ because can not sell quickly (lead to price down)
      - 慢慢 sell [during period, stock price might ↓]

Calculate Multiples

Multiples

- P/E Ratio = Price per Share/EPS
- Market Value to LTM EBITDA Ratio
- Market Value to LTM EBIT Ratio
- Market Value to LTM Sales Ratio

Calculate Multiples

- Adjusted Synergy Multiples
- Examples：
  - Market Value = $1.2 Billion
  - EBITDA = $ 0.15
  - Market Value to EBITDA Ratio = 8
  - Assume that synergy effect will increase EBITDA for $30 Million
    - » Market Value to EBITDA Ratio = 1.2/0.18 = 6.67

market doesn't think it's a good deal
[buyer 的 stock price usually ↓]
Always a premium (good for target company)
(buyers usually over-estimate the synergy)
[Actually, challenging to incorporate] culture/legal/... different.

协同效应

Summary

- Advantages of Precedent Pricing ⇒ starting point to negotiate
  - Leveraging Market Information
    - » Take Actual Acquisition Price into Consideration.
  - Timely
    - » Reflect market`s current M&A price, circumstances of capital market, and current financial situation.
  - Easiness
    - » Easy to compare with other transactions.
  - Objectiveness
    - » No need of hypothesis on future performance.

Summary

- Disadvantages of Precedent Pricing
  - Leveraging Market Information
    - » Market Sentiment may cause option pricing deviation.
  - Time Delay
    - » May not be applicable to current situation.
  - Lack of Comparable Transactions
  - Missing Transaction Details